A Conditional Reduction of the Riemann Hypothesis to Even Dominance of the Weil Quadratic Form

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Abstract

This paper presents a conditional reduction of the Riemann Hypothesis, extracted from a three-part Landscape series. The Landscape papers document the full research process, including failed routes, obstruction theorems, computational certificates, and alternative strategies. The present paper keeps only the proof-relevant chain. Following Connes' reduction of the Riemann Hypothesis to an even-dominance condition for the Weil quadratic form, we make two independent contributions. First, for the finite range $100 \leq \lambda \leq 1{,}300{,}000$, the Landscape computation reports negative interval-enclosed Galerkin gaps at 33 grid points. A 15 May 2026 audit shows that the current sine-block lower-bound construction still contains a heuristic geometric tail extrapolation; the finite-range data are therefore treated here as computer-assisted numerical evidence, not as a theorem-level CAP proof. Second, for the asymptotic regime $\lambda \to \infty$, we establish via a common Rayleigh-vector argument that $\lambda_{\min}(W^+_{\text{prime}} - W^-_{\text{prime}}) = -\Theta(\sqrt{\lambda})$ on the relevant three-mode Galerkin block. Closing the gap between this variational statement and the eigenvalue ordering $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$ for asymptotic λ requires a separate quantitative lower bound for λ_1^-; we formulate this as the *Asymptotic Variational Gap Conjecture* and outline a degenerate perturbation-theory route in Section 8. The proof of RH presented here is therefore conditional on (i) the Connes program (Connes–van Suijlekom real-zero criterion [2], which is fully proven at fixed L, together with the Hurwitz limit-passage of [1, Sec. 6.4–6.6], whose Fact 6.4 convergence is presented there as a Slepian–Pollak consequence), and (ii) the Asymptotic Variational Gap Conjecture for the odd-sector minimum eigenvalue. The Shift Parity Lemma is unconditional. The interval-arithmetic certificates as currently implemented use a heuristic geometric extrapolation for the sin-block lower bound; the 14 May 2026 v1.4 revision documents this honestly in Section 8.5.

v1.5 reformulation (14 May 2026, evening). Earlier drafts expressed the Weil quadratic form on $L^2([-L, L])$ with a real cosine/sine basis and identified the even/odd parity sectors with $t \mapsto -t$. In this formulation, the cosine and sine sectors share identical symmetric-shift auto-overlaps at every frequency $k\pi/L$, producing $\sigma(A_\lambda^+) = \sigma(A_\lambda^-) \cup \{\mu_0^+\}$ exactly and thus contradicting even dominance. A comparison against Connes–van Suijlekom's proven form [2] shows that the correct setting is $L^2([0, L])$ with the complex exponential basis $U_n = L^{-1/2} \exp(2\pi inx/L)$ and parity defined via $U_n \leftrightarrow U_{-n}$. In this setting the matrix of the Weil quadratic form has cosine- and sine-sector diagonals $D_n \mp S_n$, where $S_n = (\pi n)^{-1} \int_0^L \sin(2\pi ny/L)\, dD(y)$ captures the genuine asymmetry between the two parity sectors. Numerically $S_n > 0$ for the Weil distribution (e.g. $S_1 \approx 8.63$ at $\lambda = 1000$, giving cosine diagonal ≈ 3.26 versus sine diagonal ≈ 20.52), so even dominance is structurally present in the corrected form. Section 2 has been rewritten accordingly; downstream sections retain their legacy structure and require parallel adaptation.

The excluded gauge, total-positivity, and universal-commutator routes are not used and are referred to the Landscape series.

Keywords: Riemann Hypothesis, Weil quadratic form, even dominance, Connes program, Shift Parity Lemma, computer-assisted evidence.

1 Purpose and Relation to the Landscape Series

This paper is not a fourth installment of the Landscape series. It is a proof-only extraction. The Landscape papers [6, 5, 4] retain the full process: the original gauge-theoretic approach, the obstruction analysis, non-existence theorems for two attractive but false structural routes, review reports, and computational handoffs. Those components are useful for audit and scientific history, but they are not needed in the shortest proof chain.

The argument presented here uses the following two-part chain:

$$\begin{gathered}\text{Shift parity} + \text{IA diagnostics for } 100 \le \lambda \le 1{,}300{,}000 \\ \Longrightarrow \text{ finite-range evidence; theorem-level only under Theorem 7.1,}\end{gathered}$$

together with

$$\begin{gathered}\text{frontier dominance} + \text{Asymptotic Variational Gap Conjecture} \\ \Longrightarrow \text{ Even Dominance for } \lambda \ge \Lambda^{\dagger} \\ \Longrightarrow \text{ RH via Connes and Hurwitz (Theorem 7.3).}\end{gathered}$$

The first line is diagnostic: the Shift Parity Lemma is unconditional, while the current finite certificates still require a rigorous odd-sector lower-bound closure before they become theorem-level CAP. The second line introduces two conditional inputs: the Connes program (external) and the Asymptotic Variational Gap Conjecture (Theorem 7.2, internal to this approach; Section 8).

2 Connes' Reduction (v1.5 Reformulation)

v1.5 reformulation note. Earlier drafts of this paper (up to and including v1.4) presented the Weil quadratic form on $L^2([-L, L])$ with a real cosine/sine basis and identified the even/odd sectors with parity under $t \mapsto -t$. An audit of the operator definition in May 2026 against Connes–van Suijlekom [2] showed that the originally proven form acts on $L^2([0, L])$ with a complex exponential basis, and that the relevant parity sectors are defined via the frequency reflection $U_n \leftrightarrow U_{-n}$ on this space. Under the simplified symmetric-shift formulation of earlier drafts the cosine and sine sectors would have identical spectra (modulo the constant mode), contradicting the certified gaps; the Connes–van Suijlekom form admits a genuine asymmetry between cosine and sine sectors, captured by the quantity S_n introduced below, and it is in this form that even dominance is mathematically meaningful. The present section reformulates the setup to match the proven form; Section 8 keeps a record of the previous formulation and the diagnostic chain that led to this revision.

Hilbert space and basis. Let $L = \log \lambda$. Following [2, Section 4], take the Hilbert space

$$\mathcal{H} \;=\; L^2([0, L],\, dx),$$

with orthonormal Fourier basis

$$U_n(x) \;:=\; \frac{1}{\sqrt{L}} \exp\!\Big(\tfrac{2\pi i n x}{L}\Big), \qquad x \in [0, L],\; n \in \mathbb{Z},$$

extended by 0 outside $[0, L]$. The involution and convolution are

$$f^*(x) \;:=\; \overline{f(-x)}, \qquad (f * g)(y) \;:=\; \int_{\mathbb{R}} f(x)\, g(y - x)\, dx.$$

For U_n as above, U_n^* is supported on $[-L, 0]$ and $U_n^*(x) = L^{-1/2} \exp(2\pi i n x/L)$ on that interval.

Weil distribution. Let D be the Weil distribution on $[0, L]$,

$$D \;=\; (\log 4\pi + \gamma)\,\delta_0 \;+\; \frac{e^{y/2}}{2\sinh y}\,\chi_{(0,L]}(y)\,dy \;-\; \sum_{p,\,m\geq 1} \frac{\log p}{p^{m/2}}\,\delta_{m\log p}, \tag{1}$$

where δ_a is the Dirac mass at a and the prime sum runs over those pairs (p, m) with $m \log p \in [0, L]$, i.e. $p^m \leq \lambda$. The regularised archimedean integrand of [2, Section 4] absorbs the singularity of $e^{y/2}/(2\sinh y)$ at $y = 0$ into the diagonal modification described below.

Quadratic form. The Weil quadratic form $Q = Q_\lambda$ on trigonometric polynomials is

$$\langle f \,|\, g\rangle_Q \;:=\; \int_0^L \big[(f^* * g)(y) + (f^* * g)(-y)\big]\,dD(y). \tag{2}$$

By the identity $(f^* * g)(\pm y) = \langle f, T_{\mp y}\, g\rangle$ (with $T_u f(t) := f(t-u)$), this equals $\int_0^L \langle f, (T_y + T_{-y})g\rangle\,dD(y)$, i.e. a symmetric-shift form integrated against D. We let A_λ denote the lower-bounded self-adjoint operator on $\mathcal{H}$ associated with Q via the Friedrichs extension.

Matrix elements. A direct computation ([2, Equation 7] and Appendix below) gives, for $y \in [0, L]$,

$$(U_m^* * U_n)(y) + (U_m^* * U_n)(-y) \;=\; \begin{cases} 2(1 - y/L)\cos(2\pi n y/L), & m = n, \\ \dfrac{\sin(2\pi m y/L) - \sin(2\pi n y/L)}{\pi(n-m)}, & m \neq n. \end{cases} \tag{3}$$

The matrix of Q in the $\{U_n\}$ basis is therefore

$$\langle U_m \,|\, U_n\rangle_Q \;=\; \int_0^L \big[(U_m^* * U_n)(y) + (U_m^* * U_n)(-y)\big]\,dD(y). \tag{4}$$

Diagonal and pair-symmetric off-diagonals. Introduce

$$D_n \;:=\; \int_0^L 2(1 - y/L)\cos\!\Big(\tfrac{2\pi n y}{L}\Big)\,dD(y), \qquad S_n \;:=\; \frac{1}{\pi n}\int_0^L \sin\!\Big(\tfrac{2\pi n y}{L}\Big)\,dD(y), \tag{5}$$

for $n \neq 0$, and $D_0 := \int_0^L 2(1 - y/L)\,dD(y)$, $S_0 := 0$. Then

$$\langle U_n \,|\, U_n\rangle_Q \;=\; D_n, \qquad \langle U_n \,|\, U_{-n}\rangle_Q \;=\; \langle U_{-n} \,|\, U_n\rangle_Q \;=\; -S_n. \tag{6}$$

In particular $D_{-n} = D_n$, so the pair $\{U_n, U_{-n}\}$ acts as an invariant subspace under the matrix (4) only in the sense that its 2×2 restriction is symmetric in $\{U_n, U_{-n}\}$; full off-diagonal entries $\langle U_m | U_n\rangle_Q$ with $m \neq \pm n$ are in general nonzero but are not used in the diagonal analysis below.

Cosine–sine sector decomposition. Define the real orthonormal basis

$$\cos_n \;:=\; \tfrac{1}{\sqrt{2}}(U_n + U_{-n}), \qquad \sin_n \;:=\; \tfrac{1}{i\sqrt{2}}(U_n - U_{-n}) \qquad (n \geq 1),$$

together with U_0 for $n = 0$. Using (6),

$$\langle \cos_n \mid \cos_n\rangle_Q \;=\; D_n - S_n, \qquad \langle \sin_n \mid \sin_n\rangle_Q \;=\; D_n + S_n, \tag{7}$$

so the two parity sectors share the same diagonal coefficient D_n but differ by $\pm S_n$. The asymmetry $2S_n$ is the central object of this paper: its sign and magnitude determine whether the cosine sector or the sine sector achieves the smaller diagonal entry, and hence (modulo controlled off-diagonal corrections) which sector realises the operator's minimum eigenvalue.

Definition 2.1 (Even dominance, Connes–van Suijlekom form)**.** *Let $A_\lambda^{\cos}$ and $A_\lambda^{\sin}$ denote the Q_λ-induced operators on the cosine and sine sectors $\overline{\operatorname{span}}\{U_n+U_{-n} : n \geq 0\}$ and $\overline{\operatorname{span}}\{U_n-U_{-n} : n \geq 1\}$ respectively. We say that QW_λ has* even dominance *if*

$$\lambda_1^{\cos}(\lambda) \;<\; \lambda_1^{\sin}(\lambda) \quad \textit{with simple, isolated cosine minimum,}$$

where $\lambda_1^{\bullet}(\lambda) := \min \sigma(A_\lambda^{\bullet})$. We retain the earlier notation $\lambda_1^{\pm} = \lambda_1^{\cos/\sin}$ for compatibility with later sections.

Remark 2.2 (Numerical asymmetry of S_n). A direct computation of D_n and S_n from (5) for the Weil distribution (1) confirms that S_n is non-negligible: for $\lambda = 1000$ and $n = 1$ one finds $D_1 \approx 11.89$ and $S_1 \approx 8.63$, so the cosine-sector diagonal is $D_1 - S_1 \approx 3.26$ while the sine-sector diagonal is $D_1 + S_1 \approx 20.52$ (script `_proof-notes/verify_connes_vs_form.py`). The asymmetry $2S_1 \approx 17.27$ is of the same order as the diagonal itself; this is the mechanism by which even dominance arises in the Connes–van Suijlekom framework and was missed by the legacy $L^2([-L,L])$ formulation, where the symmetric-shift auto-overlap forces equal diagonals in the two sectors.

Downstream migration (v1.5). The above reformulation supersedes the symmetric-shift $L^2([-L,L])$ form of v1.4. Subsequent sections (Section 3 onward) were written in the legacy framework and require parallel updates: the 3×3 parity matrix $D_3(r)$ of Section 3 now corresponds to the cosine/sine asymmetry at a single shift, with $D_3(r)$ entries related to S_n contributions from a single prime; the finite certificates section requires rebuilt CAP matrices in the $\{U_n\}$ basis; the direct frontier argument carries over once C_f is re-derived as a lower bound on S_n at the frontier-prime contributions. The Shift Parity Lemma (Theorem 3.6) and the obstruction theorems NE-A, NE-B of the Landscape series remain unaffected. This migration is partial in the present revision: the new Section 2 above is the corrected operator definition, while the downstream propagation is recorded as an open task in Section 8.5. Consequently, the legacy sections below are retained as diagnostics and route documentation, not as a completed proof in the corrected U_n framework.

Theorem 2.3 (Connes–van Suijlekom real-zero criterion)**.** *Let a real even distribution define a lower-bounded self-adjoint quadratic-form operator whose lowest spectral value is simple, isolated, and has an even eigenfunction η. Then the Fourier transform $\widehat{\eta}$ has only real zeros.*

Proof. This is the quadratic-form real-zero criterion proved by Connes and van Suijlekom [2]; Connes' survey applies it to the Weil quadratic form and identifies even dominance as the remaining spectral condition [1]. ◻

Corollary 2.4 (Sufficiency of a divergent sequence)**.** *If even dominance holds for a sequence $\lambda_n \to \infty$, then the Riemann Hypothesis follows.*

Proof. For each λ_n, Theorem 2.3 gives real zeros for the corresponding truncated Fourier transform $\widehat{\eta}_{\lambda_n}$. The Hurwitz bridge of the Connes program [1, Fact 6.4 and Sections 6.4–6.6] establishes:

(a) The truncated kernels k_{λ_n} converge locally uniformly to the completed Riemann Ξ-function on every compact subset of $\mathbb{C}$, at rate $O(\lambda_n^{-1/2-\alpha})$ for some $\alpha > 0$.

(b) All k_{λ_n} and Ξ are entire functions of order one.

(c) By Hurwitz' theorem on zeros of locally uniform limits of holomorphic functions, every zero of Ξ in any open set $U \subset \mathbb{C}$ is a limit of zeros of k_{λ_n} inside U for n large.

Since each k_{λ_n} has only real zeros (Theorem 2.3), the zeros of Ξ are real as well. The non-trivial zeros of $\zeta(s)$ correspond bijectively to zeros of Ξ, hence they lie on the critical line $\Re(s) = 1/2$. $\square$

Remark 2.5 (Analytical structure of the Hurwitz bridge). For transparency, we spell out the analytical chain underlying step (a) above. The truncated kernel is

$$k_\lambda(z) = \int_{-L}^{L} K_\lambda(t)\, e^{izt}\, dt, \qquad L = \log \lambda,$$

where K_λ is the even Weil spectral kernel restricted to $[-L, L]$. As a finite Fourier integral over a bounded interval, k_λ belongs to the Paley–Wiener class PW_L: it extends to an entire function satisfying $|k_\lambda(z)| \le C_\lambda\, e^{L|\Im z|}$.

A self-contained proof of locally uniform convergence $k_\lambda \to \Xi$ would require three ingredients:

(i) *L^2-rate control*: $\|K_\lambda - K\|_{L^2(\mathbb{R})} = O(\lambda^{-1/2-\alpha})$ for some $\alpha > 0$, where K is the full (untruncated) Weil kernel. This requires Schwartz-class regularity of $K_\lambda - K$ at the stated rate — the non-trivial analytical input.

(ii) *Locally uniform convergence from L^2 and Paley–Wiener*: on any compact $\mathcal{K} \subset \mathbb{C}$, PW membership of k_λ gives $|k_\lambda(z)| \le e^{LR}\|K_\lambda\|_{L^2}$ (with $R = \max_{z\in\mathcal{K}} |\Im z|$), so the family is locally bounded; normality (Montel's theorem) and L^2-convergence then yield locally uniform convergence on $\mathcal{K}$.

(iii) *Order one*: both k_λ and Ξ have order one, as required for Hurwitz' theorem to transfer all zeros.

Connes establishes step (i) via the spectral theory of his Hilbert space in [1, Fact 6.4 and Sections 6.4–6.6]; steps (ii) and (iii) are classical. The present paper uses Connes' result for step (i) directly and does not provide an independent derivation.

Remark 2.6 (Conditioning and hypotheses). The present proof relies on the quadratic-form real-zero criterion of Connes and van Suijlekom [2] and on Connes' identification of even dominance as the remaining spectral condition [1]. Both results are currently available as arXiv preprints. Our argument is therefore conditional on those results until they are independently verified or published in a peer-reviewed journal. The hypotheses of Theorem 2.3 require: (i) that A_λ be bounded below, which holds because the archimedean kernel is integrable and the prime sum converges absolutely; (ii) that the lowest spectral value be simple and isolated — this is the content of the even-dominance condition proved below; and (iii) locally uniform convergence of the Fourier transforms of the truncated eigenfunctions, established in [1, Section 4].

3 The Parity Matrix of a Single Shift

v1.5 reformulation note. This section, originally written in the legacy $L^2([-L, L])$ cosine/sine framework, has been retained in its legacy form below (Subsection 3.2) and augmented with the Connes–van Suijlekom analog (Subsection 3.1). In the legacy framework the matrix $D_3(r)$ encodes the cos–sin parity difference at a single shift directly; in the Connes–van Suijlekom framework the analog is the single-shift contribution to the asymmetry coefficient S_n introduced in (5). Both objects encode the same content: a single prime shift, after symmetrisation in the Weil form, contributes *differently* to the cosine and sine sectors, with the difference quantified by a single trigonometric expression in the normalised shift $r = u/L$. The legacy formulation is preserved for continuity of notation in Sections 4–6, which still read in legacy form; the migration of those sections is recorded as an open task in Section 8.5.

3.1 Connes–van Suijlekom single-shift asymmetry

In the corrected setup of Section 2, a single Dirac mass at $y \in (0, L)$ with weight w in the Weil distribution D contributes to the diagonal matrix elements as

$$\Delta D_n \;=\; w \cdot 2(1 - y/L)\cos(2\pi ny/L), \qquad \Delta S_n \;=\; \frac{w}{\pi n}\sin(2\pi ny/L) \quad (n \geq 1).$$

The Cos- and Sin-sector diagonal contributions are therefore $\Delta D_n - \Delta S_n$ and $\Delta D_n + \Delta S_n$ respectively (cf. (7)), and the parity asymmetry from this single shift at mode n is

$$-2\Delta S_n \;=\; -\frac{2w}{\pi n}\sin(2\pi ny/L) \quad \text{(cosine-minus-sine diagonal contribution)}. \tag{8}$$

Lemma 3.1 (Single-shift asymmetry; v1.5 analogue of Shift Parity)**.** *Let $r := y/L \in (0, 1]$ be the normalised shift. For a single Dirac mass δ_y with the Weil prime-sign weight $w = -(\log p)/p^{m/2}$ (i.e. $w < 0$), the single-shift cosine-minus-sine asymmetry at mode $n \geq 1$ is*

$$-2\Delta S_n \;=\; \frac{2\log p}{\pi n \cdot p^{m/2}}\sin(2\pi nr).$$

This is strictly negative *(i.e. favours the cosine sector) iff $\sin(2\pi nr) < 0$, which holds on the union of intervals*

$$r \;\in\; \bigcup_{k=0}^{n-1}\Big(\tfrac{k}{n} + \tfrac{1}{2n},\; \tfrac{k+1}{n}\Big).$$

In particular, for $n = 1$ the favourable range is $r \in (\frac{1}{2}, 1)$, i.e. $\sqrt{\lambda} < p^m \leq \lambda$; this is the analogue of the legacy "Frontier" regime, except that the favourable range is the half-frontier $r > 1/2$ rather than the neighbourhood of $r = 1$.

Proof. Direct substitution of $w = -(\log p)/p^{m/2}$ into (8) gives the displayed formula. The sign condition $-2w\sin(2\pi nr) < 0$ becomes $\sin(2\pi nr) < 0$ since $w < 0$, whose solution set on $r \in (0, 1]$ is the union of intervals displayed. The frontier identification follows from $y = m\log p$ and $r = y/L = m\log p/\log\lambda$. ◻

Remark 3.2 (Multi-mode coverage). For any $r \in (0, 1)$ that is irrational with respect to $\mathbb{Q}$, the union $\bigcup_n \{n : \sin(2\pi nr) < 0\}$ is infinite; in particular, for every r outside a measure-zero set, the single shift contributes favourably to infinitely many modes. The aggregate over modes is not simply additive in eigenvalue space, but the Cesaro-style averaging of $-2\Delta S_n$ over n matches the classical Walfisz-type Mertens estimates entering the asymptotic analysis of Sections 5–6 (in their v1.5 reformulation).

Lemma 3.3 (L-independence of single-shift asymmetry)**.** *The quantity $-2\Delta S_n$ depends on the normalised shift $r = y/L$ and the mode n, but not on L separately, up to the overall prime-weight factor $\log p/p^{m/2}$ which is determined by the shift location $y = m\log p$ and the choice of $L = \log\lambda$.*

Proof. Immediate from (8): the factor $\sin(2\pi nr)$ depends only on r. ◻

3.2 Legacy parity matrix (retained for continuity)

Note. The remainder of this section is the original $L^2([-L, L])$ Cos/Sin-basis formulation that supplied the technical content of Sections 4–6. It is retained here in legacy form for backward reference; see Section 3.1 for the Connes–van Suijlekom analog. Translating downstream sections to the Connes–van Suijlekom basis is recorded as an open task in Section 8.5.

The legacy construction begins with a finite algebraic fact. Let

$$\varphi_0(t) = \frac{1}{\sqrt{2L}}, \qquad \varphi_n(t) = \frac{\cos(n\pi t/L)}{\sqrt{L}}\ (n \geq 1), \qquad \psi_n(t) = \frac{\sin((n+1)\pi t/L)}{\sqrt{L}}$$

denote the $L^2([-L, L])$-normalized even and odd basis functions. For a normalized shift $r = s/L$ define the $N \times N$ difference matrix

$$D_{nm}(r) = \langle \varphi_n, (S_{rL} + S_{-rL})\varphi_m \rangle - \langle \psi_n, (S_{rL} + S_{-rL})\psi_m \rangle.$$

Lemma 3.4 (L-independence (legacy)). *$D_N(r)$ depends only on r, not on L separately.*

Proof. Substitute $x = t/L$ in each overlap integral. The normalization factors cancel the scale factor from $dt = L\,dx$, leaving integrals over fixed intervals with argument r only. □

Proposition 3.5 (Closed diagonal entries). *For $L = 1$ the diagonal entries satisfy, for $n \geq 1$:*

$$D_{nn}(r) = (2-r)\big[\cos(n\pi r) - \cos((n+1)\pi r)\big] - \frac{\sin(n\pi r)}{n\pi} - \frac{\sin((n+1)\pi r)}{(n+1)\pi}.$$

For $n = 0$, a direct overlap-length computation gives

$$D_{00}(r) = (2-r)(1 - \cos \pi r) - \frac{\sin \pi r}{\pi}.$$

Theorem 3.6 (Shift Parity Lemma). *For every $N \geq 3$ and every $r \in (0, 2)$,*

$$\lambda_{\min}(D_N(r)) < 0.$$

Proof. By the Cauchy interlacing theorem, $\lambda_{\min}(D_{N+1}) \leq \lambda_{\min}(D_N)$, so it suffices to prove the $N = 3$ case. The trace telescopes via Theorem 3.5:

$$\mathrm{Tr}(D_3(r)) = (2-r)(1 - \cos 3\pi r) - \frac{2\sin \pi r}{\pi} - \frac{\sin 2\pi r}{\pi} - \frac{\sin 3\pi r}{3\pi}.$$

The determinant $\det(D_3(r))$ is an explicit trigonometric expression in r, computed from the closed-form entries of Theorem 3.5. We use a two-case argument.

Case 1 ($\det D_3(r) < 0$, approximately 93.3% of $(0, 2)$): the eigenvalues have mixed signs, so $\lambda_{\min} < 0$.

Case 2 ($\det D_3(r) \geq 0$, the interval $R_2 \approx [0.597, 0.729]$): in this region, $\mathrm{Tr}(D_3(r)) < 0$ ($\max_{R_2} \mathrm{Tr} \approx -0.007$). Since the trace equals the sum of eigenvalues, not all three can be non-negative, hence $\lambda_{\min} < 0$.

Completeness. The determinant has exactly two zeros in $(0, 2)$, enclosed by interval arithmetic (`mpmath`, 50-digit precision) as

$$r_1^{\det} \in [0.59651, 0.59653], \qquad r_2^{\det} \in [0.72928, 0.72930].$$

The trace zeros are similarly enclosed:

$$r_2^{\mathrm{Tr}} \in [0.58760, 0.58762], \qquad r_3^{\mathrm{Tr}} \in [0.73023, 0.73025].$$

The interleaving

$$r_2^{\mathrm{Tr}} < r_1^{\det} < r_2^{\det} < r_3^{\mathrm{Tr}}$$

holds rigorously (interval separation > 0.008 at each adjacent pair), ensuring $R_2 \subset \{\mathrm{Tr} < 0\}$. Cases 1 and 2 cover $(0, 2)$ without gap. See [5] for the verification scripts. □

Remark 3.7. The lemma is independent of the zeta function. Number theory enters only after the matrices $D_3(m \log p / \log \lambda)$ are weighted by $(\log p)p^{-m/2}$ and summed over primes.

4 Finite Certificates

v1.5 status of the CAP construction. The 33 certificates described below were computed in the legacy $L^2([-L, L])$ cosine/sine Galerkin basis with symmetric-shift operators. Within that basis, both the cosine-sector upper bound on λ_1^+ (via interval arithmetic) and the sine-sector "lower bound" on λ_1^- (via heuristic geometric extrapolation of column norms) were treated as rigorous and produced a strictly negative `cert_gap` at every grid point. The v1.5 audit (Section 8.6; companion notes `KO_CAP_AUDIT.md` and `KO_CAP_CONNESVS_RESULTS.md`) shows two structural issues with this construction:

(1) The cosine–sine equality of symmetric-shift auto-overlaps at each frequency makes $\sigma(A^+) = \sigma(A^-) \cup \{\mu_0^+\}$ exactly in the legacy operator form, contradicting the certified gaps. This is resolved in the corrected Connes–van Suijlekom form of Section 2 (in which the cosine and sine sectors of the operator have genuinely distinct diagonals $D_n \mp S_n$ via the frequency-reflection $U_n \leftrightarrow U_{-n}$), but it invalidates the literal interpretation of the certificates as proving the original $\lambda_1^+ < \lambda_1^-$ on $L^2([-L, L])$.

(2) The sine-block "lower bound" is not mathematically rigorous: it is computed as the smallest Galerkin eigenvalue minus a heuristic geometric extrapolation of off-block column norms. A rigorous lower bound would require Lehmann–Maehly, Kato–Temple with an a priori lower bound on λ_2, or a quantitative Schur–complement argument; an analysis of the obstructions is given in `KO_LEHMANN_ANALYSIS.md`.

The certificates therefore stand as *computer-assisted numerical evidence* for even dominance, consistent with the corrected Connes–van Suijlekom structural picture, but not as a fully rigorous CAP-style proof in the formal sense of computer-assisted proofs in analysis. A v1.5 reformulation of the CAP in the $\{U_n\}$ basis with a rigorous lower-bound technique is recorded as an open task.

Let

$$\Delta(\lambda) = \lambda_1^+(\lambda) - \lambda_1^-(\lambda).$$

The Landscape computation supplies interval-arithmetic certificates for 33 values of λ in the interval

$$100 \leq \lambda \leq 1{,}300{,}000.$$

Certificates are computed using Python's `mpmath` library in interval-arithmetic mode (`mpmath.iv`, 50-digit precision). The even sector uses a 4×4 Galerkin matrix ($N_{\cos} = 4$); the odd sector uses $N_{\sin} = N_0 \in [40, 90]$ (see Theorem 4.1 below). Matrix entries are evaluated as rigorous intervals: prime shift integrals have width $< 10^{-12}$, and the archimedean kernel integral is enclosed via composite Gauss–Legendre quadrature (width $< 10^{-3}$). If the sine-block lower-bound construction is replaced by a rigorous bound, a negative reported gap would imply even dominance; at present it is recorded as diagnostic numerical evidence.

Remark 4.1 (Galerkin-asymmetry justification)*.* The asymmetric truncation $N_{\cos} = 4$ vs. $N_{\sin} \geq 40$ is justified by the structural difference between the two sectors observed in [5, Section 2.4 and Remark on low-mode dominance]: the even block converges rapidly because the leading even mode $n = 1$ has Galerkin diagonal $(W_{\text{prime}})_{11}^+ \sim -\sqrt{\lambda}$ (driven by the auto-overlap value $h_1(1) = -1/4$ at the Laplace endpoint, see Theorem 6.2), with the $k \times k$ minimum eigenvalue stabilizing at $k = 4$ to within 10^{-2} for every tested λ. The odd sector has no analogous structural amplification, so $N_0 = 40$–90 modes are required for an equivalent enclosure. Crucially, the Landscape certifier reports $\lambda_{\min}(\mathrm{QW}_{\sin}^{(N)}) > \lambda_{\min}(\mathrm{QW}_{\cos}^{(4)})$ for *all* $N \leq 90$ at every tested λ, supporting, but not yet proving, that the gap is genuine and not a truncation artifact.

Remark 4.2 (Truncation safety factor). The reported Galerkin/tail truncation error of the Landscape certifier at $\lambda = 100$ is at most 0.061 (Frobenius interval radius ≤ 0.0025 in the even block, total odd-sector tail ≤ 0.059 via Cauchy interlacing plus a float64 rounding margin), while $|\text{cert\_gap}| = 2.25$ – a safety factor $\geq 34\times$. For $\lambda \geq 200$ the diagnostic safety factor exceeds $65\times$ and grows monotonically with λ throughout the sampled range [5, Section 2.9 and certificate metadata].

Proposition 4.3 (Finite certificate diagnostic). *For every certified grid point λ_k in the Landscape table,*

$$\text{cert\_gap}(\lambda_k) < 0.$$

Moreover, the reported even-sector spectral gap is positive at every grid point. These data are numerical evidence; they become a rigorous finite bridge only after the sine-block lower-bound construction is validated or replaced in the corrected Connes–van Suijlekom framework.

Proof. This is a computer-assisted diagnostic output from the Landscape project [5]. The relevant certificate files are stored in the Landscape repository under `results/certificates/`. Representative values are:

λ	$\#\{p \leq \lambda\}$	λ_1^+ upper bound	cert_gap
100	25	−11.07	−2.25
500	95	−34.13	−7.72
10,000	1,229	−216.22	−42.37
320,000	27,608	−1220.97	−241.75
1,300,000	100,021	−2503.78	−475.83

All reported intervals have negative gap. The same certificate set gives a positive intra-even gap, supporting the absence of even-sector level crossing at the sampled points. □

Remark 4.4 (Certificate structure). Each diagnostic certificate for a given λ_k consists of: (a) the $N_{\cos} \times N_{\cos}$ even-sector Galerkin matrix with interval-arithmetic entries; (b) the $N_{\sin} \times N_{\sin}$ odd-sector Galerkin matrix; (c) a rigorous eigenvalue enclosure $[\underline{\lambda}_1^+, \overline{\lambda}_1^+]$ for the minimum even eigenvalue; (d) a candidate lower bound $\underline{\lambda}_1^-$ for the minimum odd eigenvalue, currently relying on the heuristic tail extrapolation described above; (e) the reported gap $\overline{\lambda}_1^+ - \underline{\lambda}_1^- < 0$. For instance, at $\lambda = 100$: the 4×4 even matrix has minimum eigenvalue in $[-11.08, -11.06]$, the odd diagnostic lower-bound output lies above -8.82, giving reported gap < -2.24. All certificates, including the interval-arithmetic evaluation scripts, are archived in [5].

5 Frontier Dominance

v1.5 reformulation summary. In the Connes–van Suijlekom form of Section 2, the analog of frontier dominance is the asymptotic positivity of the cos/sin sector asymmetry S_n for each mode n. For $n = 1$, an Abel–summation and Laplace asymptotic computation gives

$$S_1(\lambda) \sim \frac{8\sqrt{\lambda}}{\log \lambda} \quad (\lambda \to \infty), \tag{9}$$

with the more accurate finite-L form $S_1(\lambda) \approx 8\sqrt{\lambda}/[L(1 + 16\pi^2/L^2)]$ where $L = \log \lambda$. Numerically verified to within 5−10% on $\lambda \in \{100, 300, 1000\}$ against the script `verify_connes_vs_form.py`, see `KO_FRONTIER_SN_ASYMPTOTIC.md` for the derivation.

In this v1.5 form the cosine–minus–sine diagonal gap at mode $n = 1$ is

$$\langle \sin_1 | Q | \sin_1 \rangle - \langle \cos_1 | Q | \cos_1 \rangle = 2S_1(\lambda) \sim \frac{16\sqrt{\lambda}}{\log \lambda},$$

i.e. a logarithmically shrinking fraction of the leading $\sqrt{\lambda}$ scale. This is positive asymptotic evidence for even dominance on the $O(\sqrt{\lambda}/\log\lambda)$ scale, but the eigenvalue-ordering statement still requires the Asymptotic Variational Gap or an equivalent lower-bound closure. The Legacy direct-frontier construction below is retained for backward compatibility with the rest of the manuscript; the analog S_1 construction can be developed in the v1.5 form by replacing $D_3(r)$ with the single-shift asymmetry function $f(r) := \sin(2\pi r)/\pi$ (cf. Theorem 3.1).

The Legacy asymptotic proof uses the fact that primes near the moving frontier $p \sim \lambda$ dominate the parity sum.

Define the low-mode cumulative difference matrix

$$D_{\mathrm{tot}}(\lambda) = \sum_{\substack{p,m\geq 1 \\ 0<m\log p/\log\lambda<2}} \frac{\log p}{p^{m/2}} D_3\left(\frac{m\log p}{\log\lambda}\right).$$

Split it into

$$D_{\mathrm{tot}}(\lambda) = D_f(\lambda) + D_n(\lambda),$$

where D_f contains the first-order shifts with $p \in [\lambda^{0.7}, \lambda]$ and D_n contains the remaining terms.

Lemma 5.1 (Common frontier Rayleigh vector)**.** *Let $e_1 = (0,1,0)^T$. For all $r \in [0.7, 1]$,*

$$e_1^T D_3(r) e_1 \leq -C_f, \qquad C_f = 0.4685.$$

Proof. By Theorem 3.5,

$$e_1^T D_3(r) e_1 = (2-r)(\cos\pi r - \cos 2\pi r) - \frac{\sin\pi r}{\pi} - \frac{\sin 2\pi r}{2\pi}.$$

Interval evaluation of this one-variable trigonometric function on $[0.7, 1]$ gives maximum < -0.4685, attained at the left endpoint. Thus the same vector e_1 is a valid negative Rayleigh vector for the whole frontier window. □

Lemma 5.2 (Frontier lower bound)**.** *For the frontier block,*

$$\lambda_{\min}(D_f(\lambda)) \leq -C_f W_f(\lambda), \qquad W_f(\lambda) = \sum_{\lambda^{0.7}\leq p\leq\lambda} \frac{\log p}{\sqrt{p}}.$$

Furthermore,

$$W_f(\lambda) = 2\sqrt{\lambda}(1-\lambda^{-0.15}) + O(1).$$

Proof. Use the fixed Rayleigh vector e_1 from Theorem 5.1:

$$\lambda_{\min}(D_f) \leq e_1^T D_f e_1 \leq -C_f \sum_{\lambda^{0.7}\leq p\leq\lambda} \frac{\log p}{\sqrt{p}}.$$

The asymptotic estimate follows by partial summation [8] from the Prime Number Theorem in the form $\theta(x) \sim x$. □

Lemma 5.3 (Non-frontier norm bound)**.** *There is an absolute constant C_n such that*

$$\|D_n(\lambda)\|_{\mathrm{op}} \leq C_n(2\lambda^{0.35} + O(\log\lambda)).$$

One may take $C_n = 2.2847$ from the interval-certified bound $\sup_{0<r<2}\|D_3(r)\|_{\mathrm{op}} \leq 2.2847$.

Proof. The first non-frontier part, $p < \lambda^{0.7}$ with $m = 1$, has total weight

$$\sum_{p \le \lambda^{0.7}} \frac{\log p}{\sqrt{p}} = 2\lambda^{0.35} + O(1)$$

by partial summation with the unconditional estimate $|\theta(x) - x| \le 0.006788\, x/\log x$ for $x \ge$ 89,967,803 [3]; for smaller x the sum is bounded by direct computation. The higher-shift terms satisfy

$$\sum_{p \le \lambda} \sum_{m \ge 2} \frac{\log p}{p^{m/2}} = O(\log \lambda),$$

with the $m = 2$ term giving the logarithmic growth and $m \ge 3$ bounded by convergent prime sums. Multiplying by the uniform operator-norm bound for $D_3(r)$ gives the claim. □

Proposition 5.4 (Legacy variational frontier dominance)**.** *There exists an explicit* Λ_* *such that for all* $\lambda \ge \Lambda_*$*,*

$$\lambda_{\min}(D_{\mathrm{tot}}(\lambda)) < 0.$$

With the constants above, Λ_* *lies below the certified endpoint* 1,300,000 *after the explicit error terms are evaluated.*

Proof. We argue via the Rayleigh quotient at the fixed common test vector $e_1 = (0, 1, 0)^T$ together with the Cauchy–Schwarz operator-norm bound (no eigenvalue additivity is invoked). The variational inequality gives

$$\lambda_{\min}(D_{\mathrm{tot}}) \ \le\ e_1^T D_{\mathrm{tot}}\, e_1 \ =\ e_1^T D_f\, e_1 + e_1^T D_n\, e_1.$$

For the frontier block, Theorems 5.1 and 5.2 give $e_1^T D_f e_1 \le -C_f W_f(\lambda) = -2C_f\sqrt{\lambda}(1-\lambda^{-0.15}) + O(1)$. For the non-frontier block, Cauchy–Schwarz yields $|e_1^T D_n e_1| \le \|e_1\|^2\, \|D_n\|_{\mathrm{op}} = \|D_n\|_{\mathrm{op}}$, which is bounded by Theorem 5.3. Combining,

$$\lambda_{\min}(D_{\mathrm{tot}}) \ \le\ -2C_f\sqrt{\lambda}(1 - \lambda^{-0.15}) + 2C_n\lambda^{0.35} + O(\log \lambda).$$

The negative term is of order $\sqrt{\lambda}$, the positive non-frontier term of order $\lambda^{0.35}$, so the right-hand side is strictly negative for all sufficiently large λ. Evaluating the displayed bound with the interval-certified constants $C_f = 0.4685$, $C_n = 2.2847$ and the explicit PNT/Mertens remainders gives an effective threshold $\Lambda_* \le$ 1,300,000; see [5, Proposition A6 (Direct Frontier-Dominance), Remark on direct-proof ingredients]. The asymptotic regime $\lambda \ge \Lambda_*$ and the finite diagnostic range $100 \le \lambda \le$ 1,300,000 of Theorem 4.3 are numerically compatible but do not yet give a theorem-level bridge without the lower-bound repair. □

Remark 5.5 (Computational improvement of Λ_*). The bound $\Lambda_* \le$ 1,300,000 above is conservative: it is the right endpoint of the certified interval. The exact Rayleigh gap

$$g(\lambda) = -C_f\, W_f(\lambda) + C_n\, W_n(\lambda)$$

(with W_f, W_n the precise frontier and non-frontier prime-sum weights, computed by exact prime enumeration) was evaluated as follows.
(i) Step-1 exhaustive check [171,329, 182,000]*:* Every integer λ in this range was checked; $g(\lambda) < 0$ throughout, with the first negative value $g(171{,}329) = -0.0052$.
(ii) Step-1000 coarse scan [182,000, 300,000]*:* All sampled values give $g \le -3.5$. In a 500-step window at most two prime-exit events can occur, each increasing g by at most $(C_f + C_n)\log p/\sqrt{p} \le 0.35$; the total upward perturbation ≤ 0.70 leaves a margin of ≥ 2.8 below zero. No unsampled integer in $[182{,}000, 300{,}000]$ can be positive.
(iii) Asymptotic regime $\lambda \ge 300{,}000$*:* $g \le -43$ throughout (coarse scan, step 10,000), consistent with the $-\Theta(\sqrt{\lambda})$ dominance of the asymptotic formula.

Combining (i)–(iii) with the finite diagnostic range [100, 1,300,000] gives numerical support for the threshold $\Lambda_* \leq 175{,}000$ in the variational model. The verification script `verify_lambda_star.py` (float64, safety margins $\gg 10^{-10}$) is archived in the companion repository at github.com/research-line/rh-even-dominance.

6 Tail Control and Lifting to the Weil Form

v1.5 reformulation note. This section, written in the legacy $L^2([-L, L])$ basis, controls the higher-mode tail contributions in that framework. In the Connes–van Suijlekom form of Section 2, the analogous task is to control the off-diagonal contributions of the matrix $\langle U_m|Q|U_n\rangle$ for $|m|, |n| > N$ where N is the Galerkin cutoff, plus the regularised archimedean integrand. The estimates below suggest the corresponding objects after replacing the relevant overlap functions by $\sin(2\pi my/L) - \sin(2\pi ny/L)$ in place of $D_3(r)$ and replacing the Schur-complement bound by one for the off-block norm of the Connes–van Suijlekom matrix. They are not used here as migrated U_n-proofs. Migration of the precise inequalities is open and is recorded with the analytical roadmap in `KO_SN_SUBLEADING.md`. In particular, the subleading asymptotic $S_n \sim 8\sqrt{\lambda}/[L(1 + 16n^2\pi^2/L^2)]$ replaces the leading $\Theta(\sqrt{\lambda})$ contribution of the legacy direct-frontier bound, giving an $O(\sqrt{\lambda}/\log\lambda)$ scale for the Cos–Sin diagonal gap.

Theorem 5.4 establishes the legacy variational statement $\lambda_{\min}(D_{\text{tot}}(\lambda)) < 0$ in the three-mode difference block. The full Weil operator A_λ acts on infinitely many modes and includes the archimedean contribution. The estimates below document why the legacy low-mode calculation was numerically stable; in v1.5 they are not a completed proof that the corrected Connes–van Suijlekom U_n operator preserves an even–odd eigenvalue gap.

Remark 6.1 (Archimedean contribution)*.* The archimedean kernel $e^{u/2}/(2\sinh u)$ in (2) is integrable on $(0, \infty)$, and the sector-independent term $-2e^{-u/2}\|f\|^2$ cancels in the even–odd difference. The remaining archimedean contribution involves shift overlap differences $D_N(u/L)$ weighted against the fixed kernel. Since $\|D_3(r)\|_{\text{op}} \leq 2.2847$ uniformly and the kernel integral is a constant independent of λ, the archimedean even–odd contribution satisfies $|\Delta_{\text{arch}}| \leq C_{\text{arch}}$ for an absolute constant. As the prime contribution is $\Theta(\sqrt{\lambda})$ by Theorem 5.2, the archimedean part is negligible for large λ.

Definition 6.2 (Auto-overlap profile)**.** *For the even basis function φ_n and normalized shift $u = \delta/L \in [0, 2]$, define $h_n(u) := \frac{1}{4}\langle\varphi_n, (S_{uL} + S_{-uL})\varphi_n\rangle$. The overlap integral evaluates via the substitution $x = t/L$ and the product-to-sum identity $\cos\alpha\cos\beta = \frac{1}{2}[\cos(\alpha-\beta) + \cos(\alpha+\beta)]$:*

$$\begin{aligned}\langle\varphi_n, (S_{uL} + S_{-uL})\varphi_n\rangle &= 2\int_{u-1}^{1} \cos(n\pi x)\cos(n\pi(x-u))\,dx \\ &= \cos(n\pi u)(2-u) + \frac{\sin(n\pi(2-u)) - \sin(n\pi(u-2))}{2n\pi} \\ &= (2-u)\cos(n\pi u) - \frac{\sin(n\pi u)}{n\pi},\end{aligned}$$

where the last step uses $\sin(n\pi(2-u)) = -\sin(n\pi u)$ for integer n. Dividing by 4:

$$h_n(u) = \tfrac{1}{2}(1 - \tfrac{u}{2})\cos(n\pi u) - \frac{\sin(n\pi u)}{4n\pi}, \qquad n \geq 1,$$

with $h_0(u) = (1 - u/2)/2$ (from $\langle\varphi_0, (S_{uL} + S_{-uL})\varphi_0\rangle = 2 - u$ by direct overlap-length computation). At the Laplace endpoint $u = 1$: $h_n(1) = (-1)^n/4$ for all $n \geq 0$. The normalization factor $\frac{1}{4}$ is chosen so that the prime-weighted diagonal satisfies $(W_{\text{prime}})_{nn} = \sum_p (\log p)\,p^{-1/2} \cdot 4h_n(\log p/L)$, consistent with the Weil form (2).

Lemma 6.3 (Leading-mode cancellation)**.** *Define the symmetrized off-diagonal overlap profiles $F_j(u) := S^-_{\rm sym}(0, j; u)$ (odd sector) and $G_j(u) := S^+_{\rm sym}(1, j'; u)$ (even sector) for the leading modes. Their closed forms are:*

$$S^+_{\rm sym}(1, 0; u) = \frac{\sqrt{2}\sin(\pi u)}{\pi}, \tag{10}$$

$$S^+_{\rm sym}(1, 2; u) = \frac{2(4\cos \pi u - 1)\sin \pi u}{3\pi}, \tag{11}$$

$$S^-_{\rm sym}(0, 1; u) = \frac{2(-2\sin \pi u + \sin 2\pi u)}{3\pi}, \tag{12}$$

$$S^-_{\rm sym}(0, 2; u) = \frac{3\sin \pi u - \sin 3\pi u}{4\pi}. \tag{13}$$

The overlap differences $\Delta_j(u) := F_j(u) - G_j(u)$ satisfy:

(a) *For the first two energy slots, the individual differences are $O(1)$, but their sum satisfies $\Delta_{\rm slot\,1}(u) + \Delta_{\rm slot\,2}(u) = -(2 + \sqrt{2})\, u + O(u^2)$.*

(b) *For $j \geq 2$: $\Delta_j(u) = O(u)$ individually.*

Proof. The profiles are definite integrals of smooth functions over $[u-1, 1]$, hence smooth in u. For example, $S^+_{\rm sym}(1, 0; u) = \langle \varphi_1, \varphi_0(\cdot - uL) + \varphi_0(\cdot + uL)\rangle$ evaluates to $\frac{1}{L\sqrt{2}}\int_{(u-1)L}^{L} \cos(\pi t/L)\, dt +$ (symmetric) $= \frac{\sqrt{2}\sin \pi u}{\pi}$ by direct integration; formulas (11)–(13) are obtained similarly. At $u = 0$, Fourier orthogonality gives $F_j(0) = G_j(0) = 0$. Differentiating (12)–(13) at $u = 0$: both sine overlaps have zero derivative (Taylor expansions start at u^2). For (12): rewrite as $-2\sin \pi u + \sin 2\pi u = 2\sin \pi u(\cos \pi u - 1)$, vanishing to second order. For (13): rewrite as $3\sin \pi u - \sin 3\pi u = 4\sin^3 \pi u$, vanishing to third order. The cosine derivatives are $G_0'(0) = \sqrt{2}$ from (10) and $G_2'(0) = 2$ from (11). Therefore $\sum_{\rm slots} \Delta_j'(0) = (0 - \sqrt{2}) + (0 - 2) = -(2 + \sqrt{2})$. Part (b) follows from the mean value theorem applied to $\Delta_j(u) = \Delta_j(u) - \Delta_j(0)$. □

Lemma 6.4 (Higher-mode decay)**.** *Let $B^\pm_{1j}$ denote the coupling between the leading even/odd mode and mode $j \geq 2$ induced by the prime shifts, and let g_j denote the spectral gap between mode j and the leading mode in the Galerkin diagonal. Then:*

(i) Coupling bound: $|B^\pm_{1j}| \leq 4\pi\sqrt{\lambda}/L \cdot (1 + O(1/L))$.

(ii) Gap bound: $g_j \geq 4\pi^2(j^2 - 1)\sqrt{\lambda}/L^2$ *for all $j \geq 2$ and $L \geq 5$.*

(iii) Sector difference: $|\Delta_{\rm high}|/D(\lambda) = O(1/L) \to 0$, *where $D(\lambda) = \Theta(\sqrt{\lambda})$ is the three-mode frontier advantage.*

Proof. (i) Coupling bound. For $j \neq 1$, orthogonality of the cosine basis on $[-L, L]$ gives $\langle \varphi_1, (S_0 + S_0)\varphi_j\rangle = 0$. The mean-value theorem yields $|\langle \varphi_1, (S_\delta + S_{-\delta})\varphi_j\rangle| \leq \pi\delta/L$. Summing over primes with Abel summation [8] and $\theta(x) = x + O(x\, e^{-c\sqrt{\log x}})$:

$$|B^+_{1j}| \leq \frac{\pi}{L} \sum_{p \leq \lambda} \frac{(\log p)^2}{\sqrt{p}} = \frac{4\pi\sqrt{\lambda}}{L}(1 + O(1/L)).$$

The same bound holds for $|B^-_{0j}|$ in the odd sector.

(ii) Gap bound. By Theorem 6.2, $h_n(1) = (-1)^n/4$. The prime sum concentrates at $p \sim \lambda$ (i.e., $u = \log p/\log \lambda \to 1$); by Abel summation with the PNT [8], the diagonal satisfies $(W_{\rm prime})_{nn} \sim 4h_n(1)\sqrt{\lambda} = (-1)^n\sqrt{\lambda}$. For even j: $h_j(1) - h_1(1) = \frac{1}{2}$, giving $g_j \geq \sqrt{\lambda}$. For odd $j \geq 3$: $h_j(1) = h_1(1) = -\frac{1}{4}$ (leading terms cancel). Expanding $h_j(u) - h_1(u)$ about $u = 1$ to second order gives $h_j''(1) - h_1''(1) = \pi^2(j^2 - 1)/4$, whence $g_j = 4\pi^2(j^2 - 1)\sqrt{\lambda}/L^2 + O(\sqrt{\lambda}/L^3)$. In particular, $c_{\rm gap} := 4\pi^2/L \geq 2.8$ for $L \leq 14$ ($\lambda \leq 1{,}200{,}000$).

(iii) Sector difference. Let $E_{\cos} := \sum_{j\geq 2}|B^+_{1j}|^2/g^+_j$ and $E_{\sin} := \sum_{j\geq 2}|B^-_{0j}|^2/g^-_j$ denote the resolvent-damped coupling energies in the even and odd sectors. Each is $O(\sqrt{\lambda})$ individually by (i)–(ii). The sector *difference* $|E_{\sin} - E_{\cos}|$ is further suppressed by the Leading-Mode Cancellation (Theorem 6.3): the first-order Taylor coefficients of the even and odd overlap differences at $u = 0$ sum to $-(2+\sqrt{2})$, so the leading contributions cancel pairwise. The coefficients $c_{\sin}(L) - c_{\cos}(L) = O(1/L^2)$, and therefore $|\Delta_{\text{high}}|/D(\lambda) = O(1/L) \to 0$. □

Remark 6.5 (Cancellation region for the sector difference). The cancellation used in part (iii) of Theorem 6.4 via Theorem 6.3 is a Taylor identity at $u = 0$, whereas the prime sum that drives $D(\lambda)$ is dominated by frontier shifts with $u = \log p/\log\lambda \in [0.7, 1]$. At the Laplace endpoint $u = 1$ the relevant auto-overlap values are $h_n(1) = (-1)^n/4$ (cf. Theorem 6.2), so the even and odd sectors are *anti-correlated* rather than cancelling. The $O(1/L)$-suppression claimed in part (iii) therefore relies on the averaging of the Taylor coefficient $c_{\cos}(L) - c_{\sin}(L)$ across the slot structure described in Theorem 6.3(a), not on a pointwise cancellation at frontier shifts. A direct frontier-endpoint cancellation identity in the spirit of Theorem 6.3 but anchored at $u = 1$ would strengthen this estimate; in its absence, the bound should be regarded as relying on the averaged Taylor structure of the slot decomposition and is quantitatively conservative.

Lemma 6.6 (Galerkin truncation error)**.** *The even sector is dominated by the frontier mode ($n = 1$, diagonal $\sim -\sqrt{\lambda}$); modes $n = 0, 2, 3$ contribute at lower order, so $N_{\cos} = 4$ suffices (Theorem 4.1). The odd sector lacks this structural amplification and is enclosed with $N_0 = N_{\sin} \in [40, 90]$. By the Schur-complement formula [9], the eigenvalue perturbation from modes $j > N_0$ is bounded by*

$$|\lambda_1^{\pm}(\text{full}) - \lambda_1^{\pm}(N_0)| \;\leq\; \frac{\|B_{\text{tail}}\|^2_{\text{op}}}{g_{N_0+1} - \|B_{\text{tail}}\|_{\text{op}}}.$$

Substituting Theorem 6.4(i)–(ii) gives $\|B_{\text{tail}}\|_{\text{op}} \leq 8\pi\sqrt{\lambda}/L$ and $g_{N_0+1} \geq 4\pi^2 N_0^2\sqrt{\lambda}/L^2$, so

$$|\lambda_1^{\pm}(\text{full}) - \lambda_1^{\pm}(N_0)| \;\leq\; \frac{16\sqrt{\lambda}}{N_0^2}\,(1+o(1)) \quad \textit{as } \lambda \to \infty.$$

At every sampled grid point this analytical bound is dominated by the reported Landscape safety factor of Theorem 4.2 ($\geq 34\times$ at $\lambda = 100$, $\geq 65\times$ for $\lambda \geq 200$), which is a consistency check for the reported gap, not a lower-bound proof.

Proof. The Schur-complement bound is standard [9]. Theorem 6.4(i) gives $\|B_{\text{tail}}\|_{\text{op}} \leq 8\pi\sqrt{\lambda}/L$. Theorem 6.4(ii) gives $g_{N_0+1} \geq 4\pi^2 N_0^2\sqrt{\lambda}/L^2 \gg \|B_{\text{tail}}\|_{\text{op}}$ for $N_0 \geq 40$, hence the denominator is dominated by g_{N_0+1}. The displayed bound $16\sqrt{\lambda}/N_0^2$ follows. The reported gap uses the sharper interval-arithmetic enclosure of [5, Section 2.9] with the diagnostic Cauchy tail bound described in Theorem 4.2. □

7 Main Theorem and Open Conjecture

v1.5 status of the main theorem. In the Connes–van Suijlekom form of Section 2, the main theorem statement (RH conditional on the Connes program and the Asymptotic Variational Gap Conjecture) is unchanged. What changed is the precise formulation of the conditioning:

- *External preprint input from Connes 2026 §6.5 (Fact 6.4):* the Fourier transform of the educated-guess eigenfunction k_Λ converges to Ξ uniformly on substrips $|\Im z| < \delta < 1/2$ at rate $O(\Lambda^{-1/2-\delta})$. This is stated in [1] as a consequence of the classical Slepian–Pollak prolate spheroidal wave function theory ([Connes2026, Theorem 1 of reference 47]), not as a conjecture.

- *External preprint input from Connes–van Suijlekom 2025 [2]:* the fixed-Λ quadratic-form real-zero criterion, proved within that preprint.
- *Internal open question:* simplicity and parity of the smallest eigenvalue of QW_Λ (Connes 2026 §6.6 "remaining steps" item (i)) — this is exactly even dominance as defined in Definition 2.1, addressed in the present paper.
- *Internal open question:* approximation quality $k_\Lambda \approx \xi_\Lambda$ (Connes 2026 §6.6 "remaining steps" item (ii)) — this is the Mass-Concentration question MS2 addressed in the companion Zookeeper paper [7].

The conditioning in v1.5 is therefore that *even dominance for QW_Λ* holds along a sequence $\Lambda_n \to \infty$, together with the Connes–van Suijlekom preprint criterion, gives RH via the Fact 6.4 bridge stated by Connes 2026. The companion Zookeeper question is independent and concerns whether the specific educated-guess approximant produces additional structural control.

7.1 Certified Range: Finite Even Dominance

Theorem 7.1 (Finite even dominance under lower-bound validation)**.** *Assume that the finite certificate matrices are recomputed in the corrected Connes–van Suijlekom framework and that the quantities reported as $\underline{\lambda_1^-}(\lambda)$ in Theorem 4.3 are validated as rigorous odd-sector lower bounds at each grid point. Then for every λ in that grid (33 values spanning* $100 \le \lambda \le 1{,}300{,}000$*),* QW_λ *has a simple even ground state, i.e. $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$ and the even minimum is isolated.*

Proof. Under the additional lower-bound validation hypothesis, the interval-arithmetic Galerkin diagnostics of Theorem 4.3 produce, at each λ in the grid, a rigorous upper bound $\overline{\lambda_1^+}(\lambda)$ and a rigorous lower bound $\underline{\lambda_1^-}(\lambda)$ from the even and odd Galerkin blocks respectively, together with validated tail bounds. The resulting gap $\overline{\lambda_1^+}(\lambda) - \underline{\lambda_1^-}(\lambda) < 0$ at every grid point directly establishes $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$ on the grid. Simplicity follows from the validated intra-even gap $\lambda_2^+ - \lambda_1^+ > 0$ at each grid point. □

7.2 Asymptotic Regime: Variational Statement and Open Conjecture

The asymptotic argument of Theorem 5.4 establishes the *variational* statement

$$\lambda_{\min}(W^+_{\mathrm{prime}}(\lambda) - W^-_{\mathrm{prime}}(\lambda)) \;=\; -\Theta(\sqrt{\lambda}) \tag{14}$$

on the three-mode Galerkin block, with explicit threshold $\Lambda_* \le 175{,}000$ (Theorem 5.5). This is the statement that the difference $W^+ - W^-$ has a negative minimum eigenvalue with quantitative rate of growth.

The variational direction gap. Statement (14) is *not* the same as $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$. The Rayleigh inequality $\lambda_{\min}(W^+ - W^-) \le v^T(W^+ - W^-)v$ gives an *upper* bound on the minimum of the difference, equivalently it shows the existence of a vector v with $v^T W^+ v < v^T W^- v$. For even dominance one needs, in addition, a quantitative *lower* bound on $\lambda_1^-(\lambda)$ that places the odd minimum above the certified upper bound for λ_1^+. The simple 2×2 example

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 10 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 0 \\ 0 & -5 \end{pmatrix}$$

shows that $\lambda_{\min}(A - B) = -5 < 0$ does not imply $\lambda_{\min}(A) < \lambda_{\min}(B)$; in fact $\lambda_{\min}(A) = 0 > -5 = \lambda_{\min}(B)$. The finite-range data support this picture but do not close the gap at theorem level until the odd-sector lower bound is validated; asymptotically a separate argument is required.

Conjecture 7.2 (Asymptotic Variational Gap Conjecture)**.** *There exist constants* $\Lambda^\dagger > 0$, $c_+ > c_- > 0$ *and an exponent* $\beta \in (0, 1)$ *such that for all* $\lambda \geq \Lambda^\dagger$*,*

$$\lambda_1^+(\lambda) \;\leq\; -c_+\sqrt{\lambda}, \qquad \lambda_1^-(\lambda) \;\geq\; -c_+\sqrt{\lambda} + c_-\sqrt{\lambda}\,L^{-\beta},$$

where $L = \log\lambda$*. In particular,* $\lambda_1^+(\lambda) < \lambda_1^-(\lambda)$ *for all* $\lambda \geq \Lambda^\dagger$*.*

Section 8 discusses what would be required to establish Theorem 7.2, sketches a proposed approach via degenerate perturbation theory on the asymmetric three-mode diagonal structure, and explains why the naive variational argument cannot be lifted directly.

7.3 Riemann Hypothesis: Conditional Statement

Theorem 7.3 (Riemann Hypothesis, conditional)**.** *Assume Theorem 7.2 (Asymptotic Variational Gap) and the Connes program, namely the quadratic-form real-zero criterion of [2] together with the Hurwitz bridge of [1, Fact 6.4 and Section 6.4–6.6]. Then all non-trivial zeros of the Riemann zeta function [10] lie on the critical line* $\Re(s) = 1/2$*.*

Proof. By Theorem 7.2, even dominance holds for every $\lambda \geq \Lambda^\dagger$, hence along a divergent sequence $\lambda_n \to \infty$. If the finite certificate lower bounds are later validated, Theorem 7.1 adds a finite-range bridge, but it is not needed for this divergent sequence. Applying Theorem 2.4 to the sequence: the Fourier transforms of the even-dominant Weil forms have only real zeros (Theorem 2.3), and the Hurwitz bridge transfers the real-zero property to the limiting Ξ-function. Therefore the non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = 1/2$. □

Remark on conditioning. Theorem 7.3 carries two distinct conditional inputs. The Connes–van Suijlekom criterion and the Hurwitz bridge are external results of the Connes program [1, 2], currently arXiv preprints; their analytical structure is discussed in Theorem 2.5. The Asymptotic Variational Gap Conjecture 7.2 is internal to the present approach and is the central open mathematical problem for closing the asymptotic half of even dominance unconditionally within this framework.

8 Status of the Asymptotic Lower Bound for λ_1^-

This section is not part of the proof of Theorem 7.3: it explains why the common-Rayleigh-vector argument of Theorem 5.4 cannot be lifted directly to even dominance in the asymptotic regime, and outlines a proposed approach – via degenerate perturbation theory on the asymmetric three-mode diagonal structure – toward establishing Theorem 7.2.

8.1 Why the Variational Bound Is Not Even Dominance

Theorem 5.4 establishes $\lambda_{\min}(D_{\mathrm{tot}}(\lambda)) < 0$ on the three-mode block, where $D_{\mathrm{tot}} = W^+_{\mathrm{prime}} - W^-_{\mathrm{prime}}$ (restricted to the 3×3 block). By the Rayleigh variational inequality this means: there exists a vector v (namely $v = e_1$) with $v^T W^+ v < v^T W^- v$. It does not follow that the smallest eigenvalue of W^+ lies below the smallest eigenvalue of W^-. Indeed, $\lambda_{\min}(W^\pm) \leq v^T W^\pm v$ holds for any unit vector v, so both bounds run in the same direction: $\lambda_{\min}(W^+) \leq e_1^T W^+ e_1$ and $\lambda_{\min}(W^-) \leq e_1^T W^- e_1$. A quantitative *lower* bound for $\lambda_{\min}(W^-)$ is needed.

8.2 Asymmetric Three-Mode Diagonal Structure

At the Laplace endpoint $u = 1$ (the frontier regime $r = \log p/\log\lambda \approx 1$, which dominates the prime sum), the asymptotic leading-order diagonal entries of $W^\pm_{\mathrm{prime}}$ on the three-mode block are

$$\mathrm{diag}(W^+_{\mathrm{prime}}) \;\sim\; \sqrt{\lambda}\cdot(+1,\,-1,\,+1), \qquad \mathrm{diag}(W^-_{\mathrm{prime}}) \;\sim\; \sqrt{\lambda}\cdot(-1,\,+1,\,-1).$$

These signs follow from the auto-overlap profile at $u = 1$: $h_n(1) = (-1)^n/4$ for even basis functions and $h_n^{\sin}(1) = (-1)^{n+1}/4$ for odd, weighted by the frontier prime sum $\sum_{p\in\text{frontier}} \log p/\sqrt{p} \sim 2\sqrt{\lambda}$.

The structural asymmetry is then visible:

- In W^+ the minimum diagonal $(-\sqrt{\lambda})$ is at index 1 and is *isolated*: the other two diagonals are $(+\sqrt{\lambda})$, an $O(\sqrt{\lambda})$-distance away. Off-diagonal couplings shift λ_1^+ by only *second-order* amounts.
- In W^- the minimum diagonals $(-\sqrt{\lambda})$ are at indices 0 and 2 and are *near-degenerate*, with the central diagonal $(+\sqrt{\lambda})$ separated by $O(\sqrt{\lambda})$. Off-diagonal couplings between the degenerate pair split them in *first* order, potentially driving λ_1^- further down.

A naive analysis stops here and yields *anti*-even-dominance: $\lambda_1^- < \lambda_1^+$, contradicting the 33 finite-range certificates. The resolution must come from a subleading order of the relevant off-diagonal coupling $b_{02} := (W^-_{\text{prime}})_{02}$ at the Laplace endpoint.

8.3 Elementary Cancellation in the Degenerate Odd Coupling

The off-diagonal coupling b_{02} in the odd sector at shift r involves the overlap integral

$$I_-(r) \;=\; \int_0^1 \sin(\pi x)\sin(3\pi(x-r))\,dx.$$

At $r = 1$ this vanishes by elementary sine orthogonality on $[0,1]$: $\sin(\theta - 3\pi) = \sin(\theta-\pi) = -\sin(\theta)$, so

$$I_-(1) \;=\; -\int_0^1 \sin(\pi x)\sin(3\pi x)\,dx \;=\; 0.$$

Differentiability of I_- at $r = 1$ gives $I_-(r) = O(1-r)$ near the endpoint. In the frontier window $r \in [0.7, 1]$, the weighted prime sum $\sum_p (\log p/\sqrt{p})\cdot I_-(r_p)$ with $\langle 1-r_p\rangle \sim 1/L$ on average should therefore satisfy

$$|b_{02}(\lambda)| \;=\; O(\sqrt{\lambda}/L).$$

If this estimate can be made rigorous with explicit constants, the degenerate splitting in λ_1^- is at most $O(\sqrt{\lambda}/L)$, which is *subleading* compared to a possible $-c_+\sqrt{\lambda}$ shift in λ_1^+ from the same frontier mechanism.

8.4 Proposed Three-Step Programme

Theorem 7.2 would follow from the following programme:

Step 1. Diagonal accounting. Compute the explicit diagonal entries of $W^{\pm}_{\text{prime}}$ on the three-mode block, separating leading frontier contributions from subleading tail and archimedean terms. This is bookkeeping without new mathematics, and should produce explicit constants $c^{\pm}_{nn}$ such that $(W^{\pm}_{\text{prime}})_{nn} = c^{\pm}_{nn}\sqrt{\lambda} + o(\sqrt{\lambda})$.

Step 2 (Off-diagonal control). Compute the trigonometric overlap integrals $I^{\pm}_{nm}(r) = \int_0^1 \varphi_n^{\pm}(x)\varphi_m^{\pm}(x-r)\,dx$ for $n, m \in \{0,1,2\}$ and $r \in [0.7, 1]$, and aggregate against the frontier prime sum. Establish the quantitative bound $|b_{02}(\lambda)| \le C\sqrt{\lambda}/L$ with explicit C.

Step 3 (Degenerate perturbation theory). Apply Bauer–Fike or a refined perturbation theorem for near-degenerate eigenvalues with controlled coupling. Obtain

$$\lambda_1^-(\lambda) \;\geq\; -\sqrt{\lambda}\,(1 + O(1/L)),$$

combine with the upper bound $\lambda_1^+(\lambda) \leq -\sqrt{\lambda}\,(1+c_+/\text{slowly-varying})$ from the common-Rayleigh argument (after rederivation in absolute terms, not through the difference), and conclude that $\lambda_1^+ < \lambda_1^-$ asymptotically.

Step 4 (Galerkin transfer). Use the existing Schur-complement machinery (Theorem 6.4 and Theorem 6.6) to transfer the three-mode statement to the full Weil operator $A_\lambda^\pm$. This is already in place modulo the issue noted in Theorem 6.5 below.

Honest assessment. Steps 1 and 2 are routine but laborious (several weeks). Step 3 is genuinely difficult: degenerate perturbation theory with sign-controlled couplings is an established discipline, but every explicit constant can become a subproblem. The relationship $|\lambda_1^+ - \lambda_1^-| = O(\sqrt{\lambda}/L)$ that this programme suggests is consistent with the numerical observation that the reported gap is roughly 20% of $|\lambda_1^+|$ – not a constant fraction of $\sqrt{\lambda}$, but a logarithmically shrinking fraction.

8.5 Update (14 May 2026): Strengthened Structural Concern after Step-1 Bookkeeping

A careful execution of Step 1 above (separating leading frontier diagonals via Laplace asymptotics, with numerical validation) has produced a stronger structural concern that the reviewer's original sketch did not anticipate. We record it honestly:

Symmetric-shift diagonal equivalence. In the basis of Section 3, the symmetric shift $S_u + S_{-u}$ applied to the cosine basis $\varphi_n(t) = \cos(n\pi t/L)/\sqrt{L}$ ($n \geq 1$) and to the sine basis $\psi_{n-1}(t) = \sin(n\pi t/L)/\sqrt{L}$ gives *identical* auto-overlap profiles

$$\langle \varphi_n, (S_u + S_{-u})\varphi_n \rangle \;=\; \langle \psi_{n-1}, (S_u + S_{-u})\psi_{n-1} \rangle \;=\; 2\,(1 - u/(2L))\,\cos(n\pi u/L),$$

for $u \in [0, 2L]$. The sine-correction terms in the asymmetric shifts cancel under symmetrization. Numerical verification (script `verify_diagonal_fast.py`) confirms agreement of the corresponding prime-summed diagonal entries to machine precision for $\lambda \in \{100, 300, 1000\}$.

Spectral consequence. Since $\langle \varphi_n, (S_u + S_{-u})\varphi_m \rangle = 0$ for $n \neq m$ by Fourier orthogonality (the sine-cross-terms cancel under symmetrization), the Weil operator A_λ is exactly diagonal in this Galerkin basis, with diagonal entries

$$\mu_n^+ = \langle \varphi_n, A_\lambda\, \varphi_n \rangle, \quad \mu_{n-1}^- = \langle \psi_{n-1}, A_\lambda\, \psi_{n-1} \rangle.$$

Combined with the auto-overlap equivalence, this gives $\mu_n^+ = \mu_{n-1}^-$ for every $n \geq 1$, both in the prime contribution and (by the same symmetric-shift argument) in the archimedean contribution. Therefore in this Galerkin model,

$$\sigma(A_\lambda^+) \;=\; \sigma(A_\lambda^-) \cup \{\mu_0^+\},$$

where μ_0^+ is the constant-mode eigenvalue. Numerically $\mu_0^+ > 0$ and large, hence not the minimum of $\sigma(A_\lambda^+)$. The conclusion is

$$\lambda_1^+(\lambda) \;=\; \lambda_1^-(\lambda) \quad \text{exactly}$$

in this Galerkin model, i.e., even dominance is *not* attained asymptotically.

Tension with the finite-range certificates. The 33 IA-certificates of Theorem 4.3 report cert_gap $:= \overline{\lambda_1^+}_{\text{CAP}} - \underline{\lambda_1^-}_{\text{CAP}} < 0$, which is incompatible with the above structural equivalence unless either the spectral-equivalence statement is wrong or the $\underline{\lambda_1^-}_{\text{CAP}}$ is not a rigorous lower bound. An audit of the certifier code `certifier_production.py` shows that $\underline{\lambda_1^-}_{\text{CAP}}$ is computed as $\lambda_1^{\text{Galerkin}}(N_{\text{big}} = 60) - \text{remaining} - \epsilon$, where remaining is a heuristic geometric extrapolation of column norms (no Lehmann–Maehly, Kato–Temple, or rigorous Schur–complement bound is applied). The Galerkin variational principle gives $\lambda_1^{\text{Galerkin}}$ as an *upper* bound on λ_1^-, not a lower bound, and subtracting a heuristic correction does not yield a rigorous lower bound without further analysis. The certificates should therefore be read as *computer-assisted numerical evidence*, not as rigorous mathematical proof, until the lower-bound construction is replaced by a method backed by a quantitative perturbation theorem.

Honest current status. Both pillars of the even-dominance argument (asymptotic frontier dominance and finite-range CAP certificates) are under structural concern as of this revision. We do not at present claim even dominance for QW_λ, neither asymptotically nor on the finite diagnostic range, unless one of the following can be established:

- the Galerkin operator in Eq. (2) differs from the actual Connes operator in a way that breaks the cosine–sine frequency symmetry; or
- the heuristic tail correction in $\underline{\lambda_1^-}_{\text{CAP}}$ can be replaced by a quantitative perturbation-theoretic lower bound, and the resulting rigorous reported `cert_gap` is still negative.

The Shift Parity Lemma (Theorem 3.6), the explicit Laplace-endpoint diagonal formulas, the non-existence theorems NE-A and NE-B (Landscape Part II), and the numerical phenomenology of the reported gap are unaffected by this concern and remain valid as independent structural results.

This update is in the spirit of treating the present manuscript as a continuously evolving draft. The companion notes `_proof-notes/K0_STEP1_FINDINGS.md` and `_proof-notes/K0_CAP_AUDIT.md` provide the full analytical and numerical record of the present concern.

8.6 Update (15 May 2026): Connes-vS Operator Form Resolves the Structural Tension; Rigorous CAP Closure Remains Open

A subsequent comparison of the operator definition against Connes–van Suijlekom [2] clarified that the strengthened K0 concern (the spectral equivalence $\sigma(A^+) = \sigma(A^-) \cup \{\mu_0^+\}$) was an artefact of the legacy operator realisation on $L^2([-L, L])$ with real cosine/sine basis under $t \mapsto -t$ symmetry, not of the actual Connes operator. The proven Connes–van Suijlekom operator acts on $L^2([0, L])$ with the complex exponential basis U_n, and the cosine/sine parity sectors are defined via the frequency reflection $U_n \leftrightarrow U_{-n}$. In this corrected formulation (Section 2 v1.5 reformulation) the diagonals $\langle \cos_n |Q| \cos_n\rangle = D_n - S_n$ and $\langle \sin_n |Q| \sin_n\rangle = D_n + S_n$ differ by the genuinely non-trivial asymmetry $2S_n$, where $S_n = (\pi n)^{-1} \int_0^L \sin(2\pi ny/L)\, dD(y)$. Direct computation (script `_proof-notes/verify_connes_vs_form.py`) gives $S_n > 0$ for the Weil distribution and all tested λ. For $\lambda = 1000$, $n = 1$: $S_1 \approx 8.63$, cos diagonal ≈ 3.26, sin diagonal ≈ 20.52. Asymptotically $S_1 \sim 8\sqrt{\lambda}/\log\lambda$ (script `verify_connes_vs_form.py`, derivation in `K0_FRONTIER_SN_ASYMPTOTIC.md`). The structural foundation for even dominance is therefore present in the correct operator form.

The remaining task is a *rigorous* computer-assisted-proof closure of $\lambda_1^{\cos}(\lambda) < \lambda_1^{\sin}(\lambda)$ in this form. A systematic analysis of the available rigorous lower-bound techniques is recorded in `_proof-notes/K0_LEHMANN_ANALYSIS.md`, with the conclusion:

- Temple's inequality $\lambda_1 \geq \mu - \eta^2/(\rho - \mu)$ requires $\rho \leq \lambda_2$ — an *a priori lower bound* on the second eigenvalue, which the Galerkin variational principle does not provide. The naive substitution $\rho = \mu_2$ *is not rigorous*: $\mu_2 \geq \lambda_2$ by Cauchy interlacing, hence using μ_2 in the denominator underestimates the correction.

- Lehmann–Maehly's generalised inequality requires $\sigma \leq \lambda_{N+1}$, again an a priori lower bound on a higher eigenvalue.
- Bauer–Fike identifies the *nearest* eigenvalue to μ, not specifically the lowest, and so is ambiguous without additional gap information.
- Operator-norm bounds give $\lambda_1 \geq -\|A\|_{\mathrm{op}}$ with $\|A_\lambda\|_{\mathrm{op}} \leq 4\sqrt{\lambda} + O(\log \lambda)$, which is far too loose.
- Schur-complement and Gerschgorin bounds require either a lower bound on the complementary block or diagonal dominance. Gerschgorin fails for the Connes–van Suijlekom Galerkin matrix because the off-diagonal row sums diverge logarithmically: the cross-overlap $\sin/(\pi(n-m))$ has only $1/|n-m|$ decay.

The common underlying obstacle is the absence of an *a priori lower bound on* λ_2 or on the tail spectrum, which Standard CAP techniques uniformly require. Three medium-term research directions to address this are sketched in `KO_LEHMANN_ANALYSIS.md`: (i) Mollification of the Weil distribution to discrete spectrum, followed by stability of the $\epsilon \to 0$ limit; (ii) Toeplitz approximation and symbol-based perturbation theory exploiting the near-Toeplitz structure of the Connes–van Suijlekom Galerkin matrix; (iii) reverse-engineering the proof technique of Connes–vS 2025 (which contains an internal Hurwitz argument and thus must implicitly handle related lower-bound questions).

Current honest position. The Connes–van Suijlekom operator form is the correct setting and exhibits even dominance numerically with consistent margin across λ and N. The asymptotic S_1-asymptotic gives a quantitative Cos–Sin diagonal gap of order $\sqrt{\lambda}/\log\lambda$. A rigorous CAP-style closure in the sense of producing a certified lower bound on $\lambda_1^{\sin}$ that strictly exceeds a certified upper bound on $\lambda_1^{\cos}$ remains open and requires a non-trivial technical advance beyond straightforward Galerkin + Temple/Lehmann. We document this honestly in line with the convention that the present manuscript is a continuously developing draft.

9 What Is Not Used

The proof does not use the following Landscape material:

- the original gauge-thermodynamic route and its obstructions;
- an absolute PF_∞ or total-positivity claim for the full prime operator;
- a universal commuting Sturm–Liouville operator;
- Hellmann–Feynman monotonicity between certificate points;
- the narrative Dungeon/Memorial presentation of excluded routes.

Those results remain valuable as route classification and audit history, but they are not part of the proof chain above.

A Constants Audit Table

All explicit numerical constants used in the proof, with their source, role, and verification mechanism. Constants flagged "IA" are interval-arithmetic certified (`mpmath.iv`, 50-digit precision); "CAP" are the 33 Landscape computer-assisted certificates; "PNT/Mertens" are explicit asymptotic remainders.

Constant	Value	Role / Source	Verification
C_f	0.4685	Common frontier Rayleigh quotient $-d_{11}(r) \geq C_f$ on $r \in [0.7, 1]$ (Theorem 5.1)	IA on $[0.7, 1]$ (1D)
C_n	2.2847	Operator-norm bound $\sup_{r\in(0,2)} \lVert D_3(r)\rVert_{\mathrm{op}} \leq C_n$ (Theorem 5.3)	IA on $(0, 2)$ (3D)
c_{gap}	$\geq 4\pi^2/L \geq 2.8$	Spectral-gap constant for $L \leq 14$ (Theorem 6.4(ii))	Laplace asymptotics
$N_{\cos}$	4	Even-block Galerkin truncation (Theorem 4.1)	Stability check $k = 3, 4, 5$
$N_{\sin}$	40–90	Odd-block Galerkin truncation (Theorem 4.1)	Cauchy tail bound
Λ_*	$\leq 175{,}000$	Asymptotic threshold; prime-sum computation (Theorem 5.5); conservative bound $\leq 1{,}300{,}000$ (Theorem 5.4)	Exact prime enum.
$\theta(x) - x$	$\leq 0.006788 \frac{x}{\log x}$	Effective Chebyshev bound for $x \geq 89{,}967{,}803$ (Dusart 2010, [3])	Unconditional
|cert_gap| at λ=100	≥ 2.25	First CAP (Theorem 4.3)	CAP
|cert_gap| at λ=1.3·10^6	≥ 475.83	Last CAP	CAP
Safety factor at λ=100	$\geq 34\times$	|cert_gap| vs. Galerkin/tail error (Theorem 4.2)	IA + Cauchy tail
Safety factor at $\lambda \geq 200$	$\geq 65\times$	Same, monotone in λ	IA + Cauchy tail
c (slot cancellation)	$2 + \sqrt{2}$	Closed-form constant (Theorem 6.3(a))	Symbolic (sympy)

The Landscape companion repository archives the verification scripts. The most relevant entries are:

- `shift_parity_cert_v3_targeted.py` – Shift Parity Lemma;
- `certifier_production.py` – the 33 CAP certificates;
- `partA_proof_sketch.py` – frontier-dominance check;
- `euler_maclaurin_certifier.py` – asymptotic threshold;

see [5, `scripts/`]. A dedicated companion repository for the present extraction is mirrored at `github.com/research-line/rh-even-dominance`.

AI Disclosure

Statement on the Use of AI Tools

This manuscript was prepared with moderate assistance from AI-based language models (Claude, Anthropic; Gemini, Google; ChatGPT, OpenAI). The use encompassed the following areas:

- *Research:* Assistance with systematic literature search, identification and summarisation of relevant research literature.

- *Structuring:* Support in outlining and logically organising the manuscript, in particular the proof-only extraction from the Landscape series.
- *Drafting:* Assistance in formulating individual sections and paragraphs based on core arguments and theses specified by the author.
- *Language:* Grammar and spell checking, linguistic revision.
- *Formatting:* Preparation and verification of citations and bibliographies; LaTeX typesetting support.
- *Internal review:* A 7-phase model-based review chain (constructive, expert, adversarial) was used to identify and address structural and citation issues prior to submission.

The entire research process – including the research question, methodological decisions, substantive argumentation, critical evaluation of sources, and conclusions – was conceived, directed, and borne by the author. All computer-assisted certificates were computed by deterministic numerical scripts independently of AI. All AI-generated suggestions were critically reviewed, revised, and where appropriate rejected. The author reviewed all content to the best of his knowledge but cannot guarantee complete independent verification of all factual claims and references.

Full scholarly and substantive responsibility for this article rests with the author, Lukas Geiger.

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